SmartMart

Online Supermarket System

Details:

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Introduction

What is SmartMart?

Project Overview

SmartMart is a feature-rich browser-based Online Supermarket System that simulates a real-world e-commerce grocery platform. It provides a seamless shopping experience for Customers and a powerful management dashboard for Staff and Admins — all in a single HTML/JS/CSS web application.

The system is AI-assisted: all source code was generated using AI tools (ChatGPT / Copilot) with targeted prompts, demonstrating AI-driven software development.

Customer Storefront

Admin Dashboard

Staff Order Manager

Smart Alerts Engine

Inventory Tracker

Profit & Loss View

Objectives

Goals we set out to achieve

01
Multi-Role Access

Separate login flows for Customer, Staff, and Admin with role-specific views and feature access control.

02
Full E-Commerce Flow

Product browsing, cart management, checkout with GST calculation, and order confirmation for customers.

03
Live Inventory Management

55 real-world Indian grocery products tracked with stock levels, reorder points, and expiry dates.

04
AI-Powered Alert Engine

Automated expiry and low-stock alerts categorised by severity (Critical, Warning, Info) for each role.

05
Analytics & Profit Reporting

Admin-only dashboards showing sales trends, category performance, P&L statements, and top products.

06
AI-Assisted Development

Entire codebase generated using AI prompting (ChatGPT / GitHub Copilot), showcasing modern AI dev workflows.

Stakeholders

Who uses SmartMart?

Customer

- ✓ Browses 55+ grocery products
- ✓ Filters by category
- ✓ Adds items to cart
- ✓ Checks out with GST billing
- ✓ Tracks order history

Staff

- ✓ Views all incoming orders
- ✓ Updates order status
- ✓ Monitors expiry alerts
- ✓ Tracks low stock alerts
- ✓ Manages inventory table

Admin

- ✓ Full access to all sections
- ✓ Profit & Loss dashboard
- ✓ Sales analytics & charts
- ✓ Supplier management
- ✓ All alerts + watch-level

Technologies Used

Programming Languages, Tools & Software

HTML5

Semantic structure, multi-view SPA layout, modals, forms

CSS3

Amazon-style navigation, card layouts, responsive grids

JavaScript (ES6)

All app logic — login, cart, orders, charts, alerts

Chart.js 4.4

Bar, pie, line, doughnut charts for analytics & P&L

ChatGPT / GPT-4

Primary AI tool used to generate all code via prompts

GitHub Copilot

In-editor AI assist for code completion and refinement

VS Code

Code editor used to organise and run the project locally

Browser (Chrome)

Runtime environment; no backend/server required

AI Prompts Used — Part 1

Prompts that generated the core codebase

P1

Login & Role System

"Build a login screen for a supermarket system. Users can enter their name, select a role (Customer, Staff, Admin) using clickable role cards, and enter any demo password. After login, show different screens based on role."

P2

Customer Storefront

"Create an Amazon-style customer shop with a top navbar (logo, search bar, cart button), category sub-nav, left sidebar category filter, and a product grid. Products should have name, price, rating, add-to-cart button."

P3

Cart & Checkout Modal

"Add a slide-in cart drawer showing selected items, quantity controls, subtotal, 5% GST, and grand total. Include a checkout modal that collects name, phone, address and places the order. Show order success with order ID."

P4

Admin Dashboard Overview

"Create an admin dashboard sidebar with sections: Overview, Alerts, Inventory, Orders, Profit & Loss, Analytics, Suppliers. The Overview should have metric cards (products, revenue, profit, low stock, expiry alerts)."

AI Prompts Used — Part 2

Prompts that generated advanced features

P5

Product Data Store

"Create a JavaScript data.js file with 55 real Indian supermarket products across 12 categories (Dairy, Fruits, Vegetables, Snacks, Beverages, Grains, Bakery, Oils, Personal Care, Frozen, Spices, Confectionery) with fields: id, name, cat, cost price, sell price, stock, sold, reorder level, expiry date, supplier, rating, reviews."

P6

Alert Engine

"Build an alerts engine that scans all products and generates alerts for: (1) expired items, (2) items expiring within 1–2 days (critical), (3) items expiring in 3–5 days (warning), (4) out-of-stock items, (5) critically low stock. Filter alerts by role — staff see expiry+stock, admin also see watch-level."

P7

Analytics & Charts

"Using Chart.js, add charts: weekly sales bar chart (revenue vs cost), category pie chart, inventory stock bar chart, hourly order volume, top 10 products bar chart, revenue by category doughnut chart, profit/loss 30-day line chart, and top margin products horizontal bar."

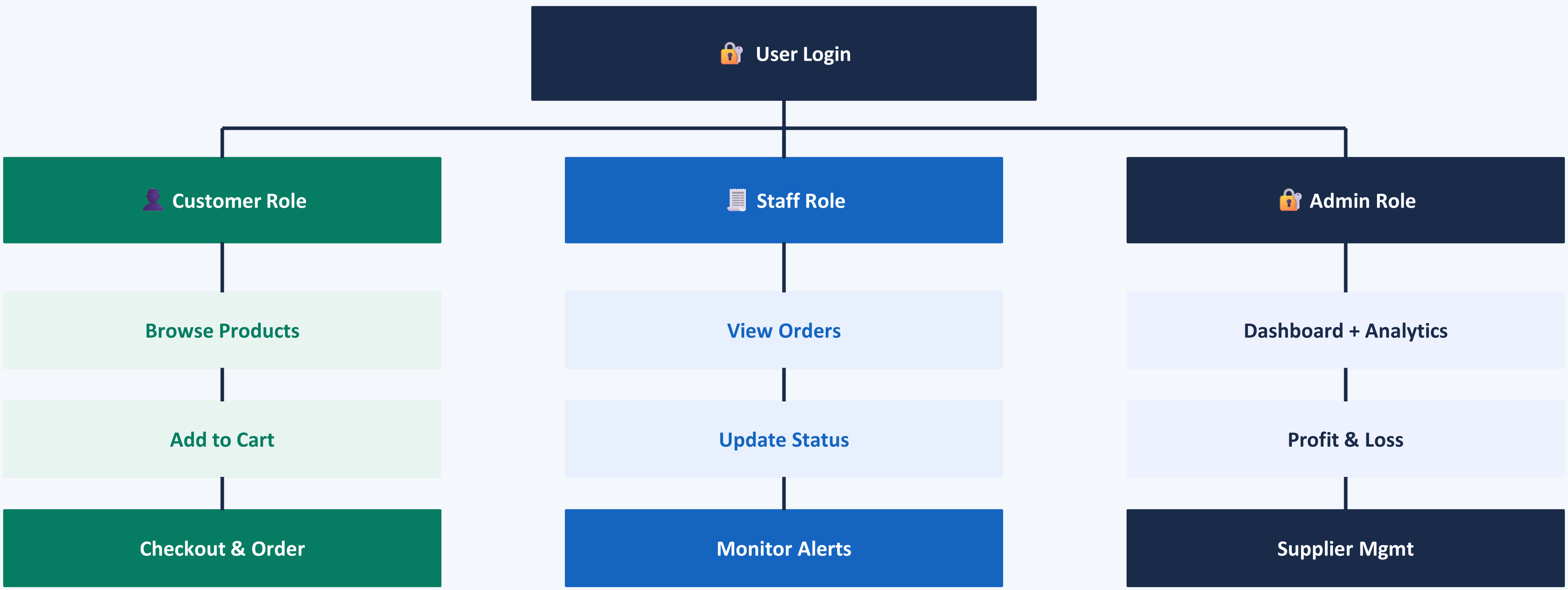
P8

Orders & Suppliers

"Create an orders management section where staff/admin can see all customer orders with status badges (Pending/Packed/Delivered), update status via dropdowns, and view order totals with GST. Also add a supplier directory table with 12 suppliers, their contact info, last order date, and status."

System Workflow

How SmartMart works end-to-end

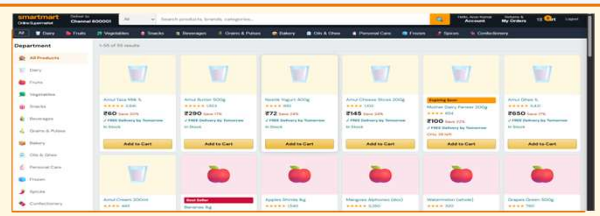


Front-End Application View

Key screens of the SmartMart web application

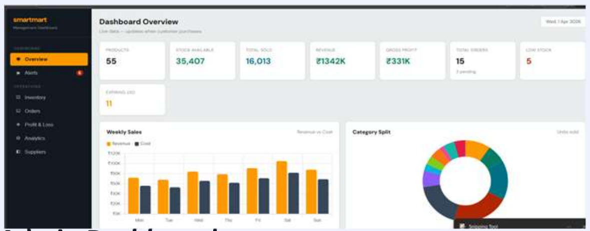
Login Screen

Role-based sign-in with Customer, Staff & Admin cards



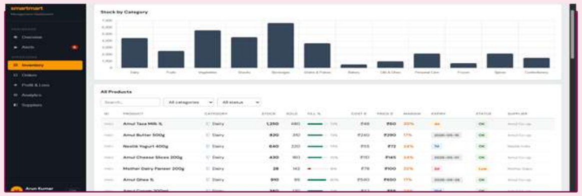
Customer Store

Amazon-style product grid with search, categories & cart



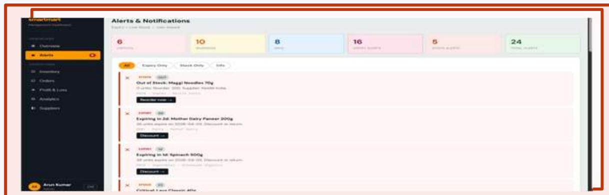
Admin Dashboard

Overview with KPI metric cards, weekly sales & category charts



Inventory Manager

55-product table with stock levels, expiry & supplier info



Alert Engine

Role-filtered critical/warning/info alerts with action buttons

Orders & Profit View

Customer order tracking + P&L, analytics charts for admin

AI Tools Used

Tools leveraged for this project

ChatGPT (GPT-4o)

Primary Code Generator

All HTML structure, JavaScript logic, CSS styling, and data files were written via detailed prompts to ChatGPT. Each feature (login, cart, alerts, charts) was built prompt-by-prompt.

GitHub Copilot

In-Editor AI Assistant

Used inside VS Code for inline code suggestions, function completions, and debugging help while organising and refining the AI-generated code.

Claude / Gemini

Research & Ideation

Used for planning the system architecture, deciding which features to include for each role, and generating the product dataset structure.

References

Sources and tools used in this project

[1] OpenAI ChatGPT — chat.openai.com — Used for AI-assisted code generation

[2] GitHub Copilot — github.com/features/copilot — In-editor AI code completion

[3] Chart.js Documentation — chartjs.org — Bar, line, pie & doughnut chart library

[4] MDN Web Docs — developer.mozilla.org — HTML5, CSS3, JavaScript ES6 reference

[5] Amazon Design System — amazon.in — UI/UX inspiration for the customer storefront layout

[6] GST India — gst.gov.in — Goods and Services Tax rate reference for billing module

[7] Statista / NielsenIQ — Market data reference for Indian supermarket product pricing

[8] VS Code — code.visualstudio.com — Code editor used for development